P8106_HW2_SY3352

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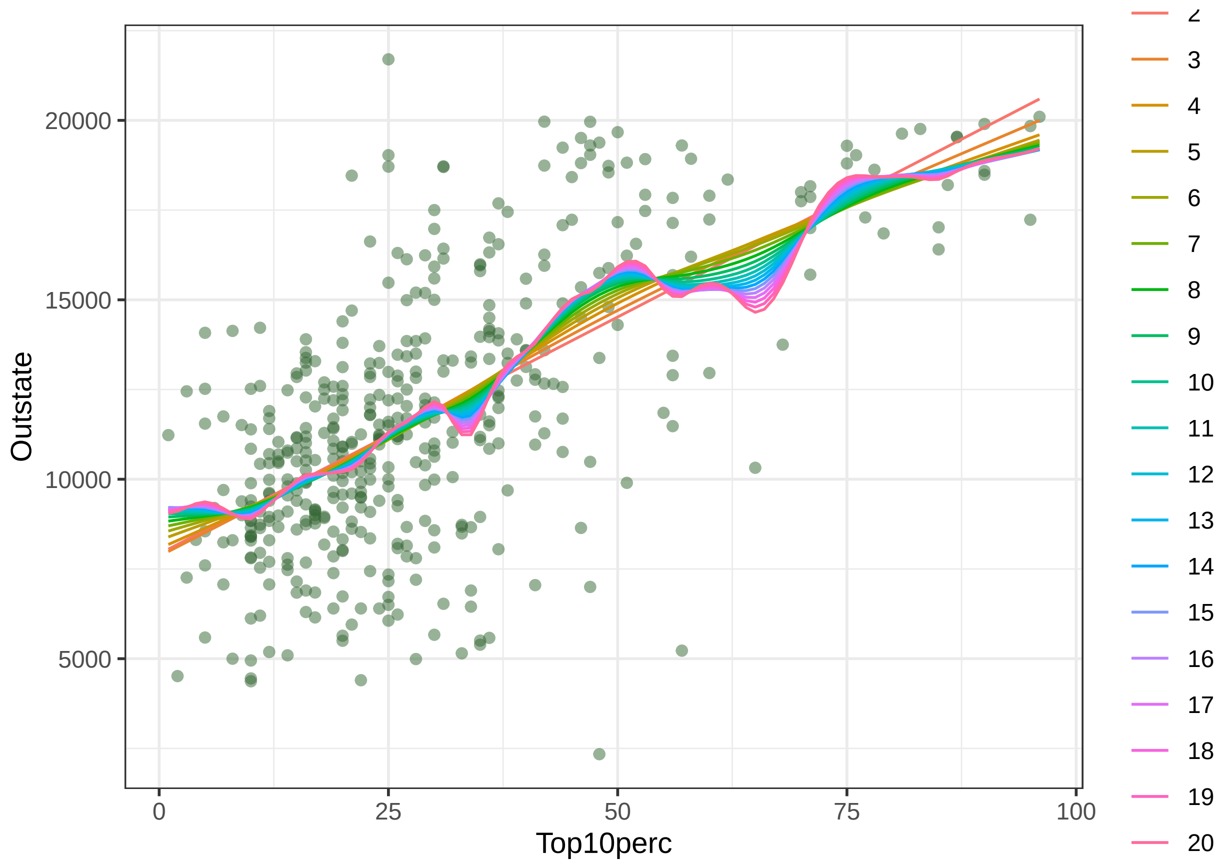
```r
library(caret)
library(tidymodels)
library(splines)
library(mgcv)
library(pdp)
library(earth)
library(tidyverse)
library(ggplot2)
```

Problem a

```r
data <- read.csv("College.csv") |> na.omit()

set.seed(2026)
data_split <- initial_split(data, prop = 0.8)

training_data <- training(data_split)
testing_data <- testing(data_split)

fit.ss <- smooth.spline(training_data$Top10perc, training_data$Outstate)

Top10perc.grid <- seq(from = min(training_data$Top10perc),
                      to = max(training_data$Top10perc), by = 1)
p <- ggplot(data = training_data, aes(x = Top10perc, y = Outstate)) +
  geom_point(color = rgb(.2, .4, .2, .5))

df_values <- seq(2, 20, by = 1)
pred_list = list()

for (df_ss in df_values) {
  fit <- smooth.spline(training_data$Top10perc, training_data$Outstate,
                       df = df_ss)
  pred <- predict(fit, x = Top10perc.grid)
  pred_list[[as.character(df_ss)]]  <- data.frame(
    Top10perc1 = Top10perc.grid, Outstate1 = pred$y, df = as.factor(df_ss))
}

pred_df_all <- bind_rows(pred_list)

p + geom_line(aes(x = Top10perc1, y = Outstate1, color = df, group = df),
              data = pred_df_all) + theme_bw() + labs(color = "df")
```

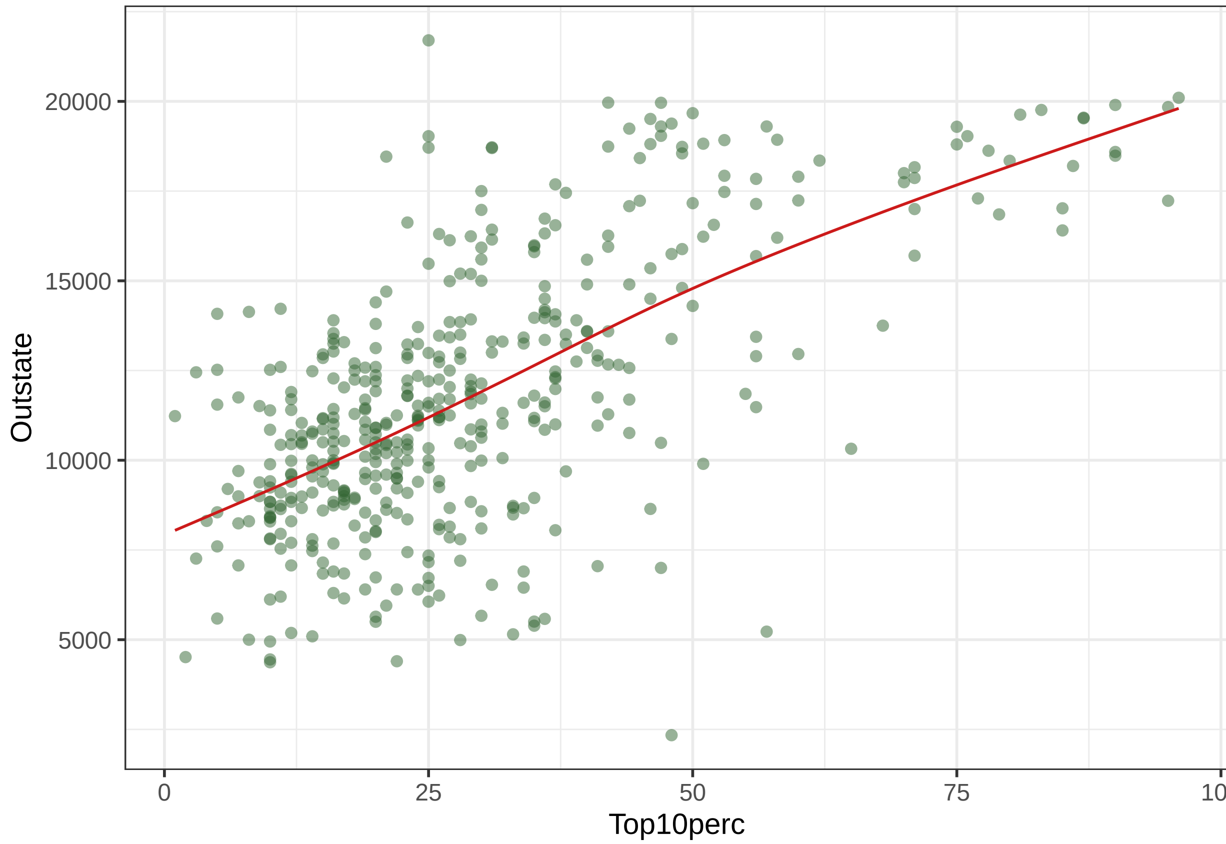


I fitted the degree of freedom from 2 to 20, it can be seen from the plot that as the degree of freedom go larger, the fitted curve become more wiggly and overfitting. With a low degree of freedom (2 to 4), the fitted curve is very smooth and almost straight, indicating that it might be underfitting and miss the non-linear pattern.

```
fit.ss$df
```

```
## [1] 3.399278
```

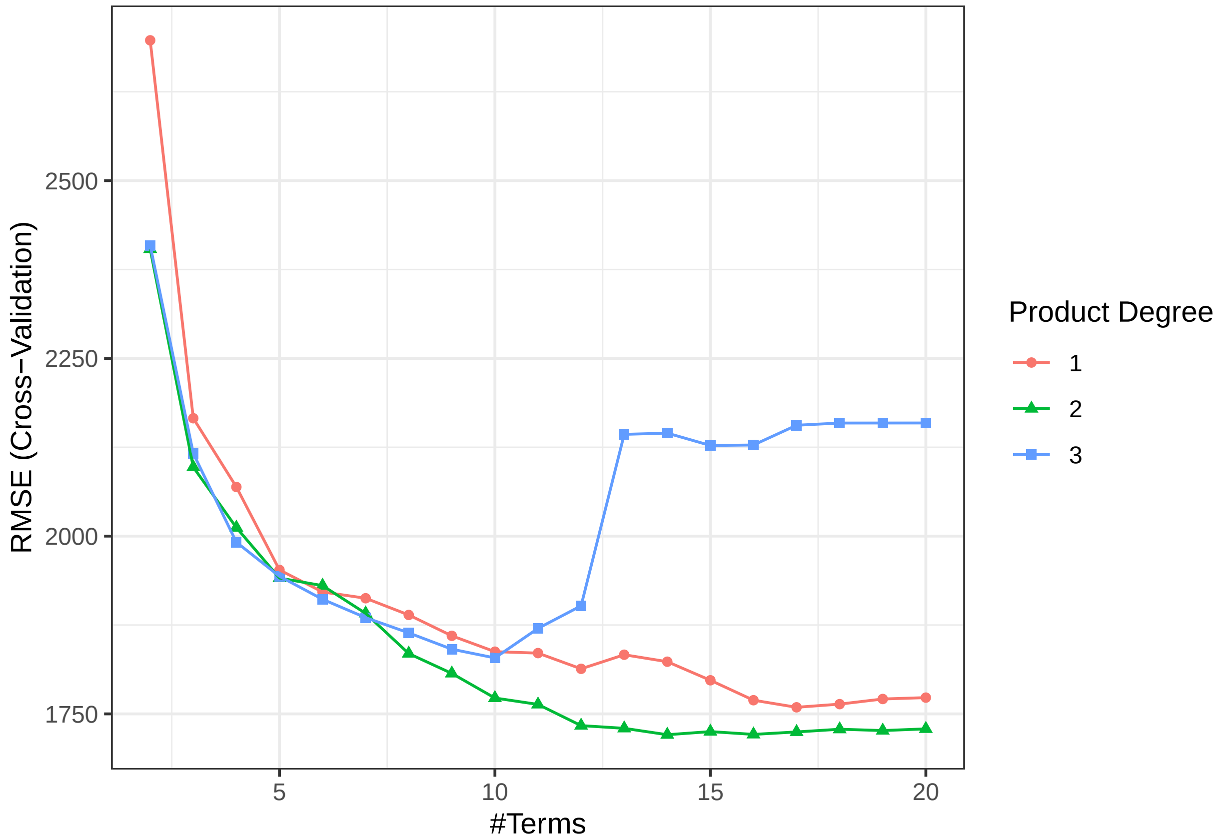
```
pred.ss <- predict(fit.ss, x = Top10perc.grid)
pred.ss.df <- data.frame(Top10perc1 = Top10perc.grid, Outstate1 = pred.ss$y)

p + geom_line(aes(x = Top10perc1, y = Outstate1),
              data = pred.ss.df,
              color = rgb(.8, .1, .1, 1)) + theme_bw()
```



The selected df is 3.3992784. Generalized cross-validation is used to select the degree of freedom because it efficiently approximates leave-one-out cross-validation error and is the standard approach for smoothing spline tuning.

Problem b

```
ctrl1 <- trainControl(method = "cv", number = 10)

mars_grid <- expand.grid(degree = 1:3, nprune = 2:20)

x_train <- model.matrix(Outstate ~ ., training_data)[, -1]
y_train  <- training_data$Outstate

x_test  <- model.matrix(Outstate ~ ., testing_data)[, -1]
y_test  <- testing_data$Outstate

set.seed(2026)
mars.fit <- train(x_train, y_train,
                  method = "earth",
                  tuneGrid = mars_grid,
                  trControl = ctrl1)

ggplot(mars.fit) + theme_bw()
```




```r
mars.fit$bestTune
```

```
##    nprune degree
## 32     14      2
```

```r
coef(mars.fit$finalModel)
```

```
##                                          (Intercept)
##                                         1.774243e+04
##                                      h(15954-Expend)
##                                        -8.733463e-01
##                                      h(83-Grad.Rate)
##                                        -3.363461e+01
##                  h(Apps-2097) * h(Room.Board-4150)
##                                         1.017795e-04
##                  h(2097-Apps) * h(Room.Board-4150)
##                                        -1.144070e-03
##                                     h(1400-Personal)
##                                         1.144294e+00
## CollegeCreighton University * h(F.Undergrad-1405)
##                                        -3.450469e+00
##                            CollegeBennington College
##                                         7.375040e+03
##            h(4150-Room.Board) * h(23-perc.alumni)
##                                        -9.544041e-02
##                                       h(2003-Accept)
##                                        -4.605971e+00
```

```
##                                     h(Enroll-828)
##                                     -1.622397e+00
##                                     h(828-Enroll)
##                                      4.264046e+00
##                  h(2306-Apps) * h(15954-Expend)
##                                      2.413358e-04
##              h(2003-Accept) * h(Room.Board-3484)
##                                      1.132795e-03
```

Problem c

Problem d

Problem e